

Henry Asante

Grambling, Louisiana (71245)

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Education

Grambling State University (GSU)

Graduating May 2028

BSc. Computer Science (GPA: 4.0/4.0)

Grambling, Louisiana

Coursework: Deep Learning, Operating Systems, Artificial Intelligence, Data Structures, Object-Oriented Design

Skills

- **Languages/Frameworks:** Java, JavaScript, Python, C/C++, React, Node, Flask, HTML, SQL, REST APIs
- **AI & Dev. Tools:** GitHub Copilot, Cursor, Claude Code, Prompt Engineering, LLM Code Review, Agentic Dev
- **Data & Infra:** MongoDB, Docker, Kubernetes, AWS, Git, CI/CD, Linux, Spark, TensorFlow, PyTorch, Pandas

Experience

Allstate Insurance Company

Northbrook, Illinois

Software Engineer Intern

May 2026 – Present

- Develop UI automation solutions using Playwright & JavaScript to validate onboarding & user journey workflows.
- Contribute to CI/CD automation using GitHub Actions & Jenkins, supporting continuous testing across dev pipelines.
- Collaborate on a production codebase using Git/GitHub, participating in PRs, code reviews, & Agile development.
- Utilize REST APIs, Linux, & AI-assisted dev to support automation, debugging, integration, & software delivery.

Dell Technologies

Round Rock, Texas

Software Engineer Intern

May 2025 – August 2025

- Boosted hardware defect regression testing speed by over 100% through an AI-driven test case recommendation system.
- Reduced hardware issue–testcase mapping errors by 10% using LLMs to analyze 1,200+ Jira tickets & 5,000+ testcases
- Achieved ~96% recommendations quality accuracy using precision/recall metrics, ensuring high-confidence test coverage
- Built user-friendly React dashboard for SQA teams to visualize AI recommendations, test coverage, & historical trends.

AmaliTech NSP Training

Remote

Software Developer Trainee

May 2024 – August 2024

- Built a Java quiz app w/ 15.6k+ Q&A, achieving 99.9% uptime by deploying Spring Boot APIs on PostgreSQL.
- Reduced API latency by ~40% via SQL query optimization, Redis caching, & multithreading for concurrent sessions
- Ensured 95%+ code coverage by writing JUnit/Mockito tests & collaborating in Agile sprints.

KNUST Engineering Lab

Kumasi, Ghana

Full Stack Engineering Intern

May 2023 – August 2023

- Built a cross-platform CAD & circuit simulation app using Vue.js, TypeScript, Python, & Electron.
- Developed frontend components & RESTful backend services connecting UI workflows with simulation engines.
- Developed reusable components & automated testing workflows, improving reliability, maintainability, & scalability.
- Integrated an AI chatbot using RAG pipelines to provide troubleshooting insights from circuit configurations.

Projects

Goal Misgeneralization in a Model-Based Agent | Python, PyTorch, MPC, CEM Planning, Simulation [view project](#)

- Engineered a physics-based simulation pipeline in a 4D continuous state space to evaluate autonomous agent robustness
- Integrated a neural dynamics model with a CEM planner for 200-step trajectory rollout and receding-horizon MPC.
- Built a reproducible evaluation framework across 4 environment shifts with up to 2× changes in mass and friction.
- Analyzed simulation telemetry, showing low next-state MSE ($\sim 10^{-3}$) despite failure from proxy reward misalignment.

Spinal Lower Back Pain Classifier | TensorFlow/Keras, MobileNet-V2, PyDICOM, Groq Llama 3.3 [view project](#)

- Deployed a full-stack Streamlit app with LLM-powered diagnostic analysis using Groq's Llama 3.3 API.
- Built and iteratively improved a CNN-based MRI classifier using MobileNet-V2, achieving ~72% training accuracy.
- Visualized accuracy & loss curves to diagnose overfitting & guide future model enhancements via fine-tuning.

Orbit-sh: A Modular Unix Shell with Systems Integration | C, Unix, Linux Systems Programming [view project](#)

- Built a Unix shell in C supporting command parsing, foreground/background execution, & built-in commands.
- Implemented pipeline support (`|`) using `pipe()` and `dup2()` for inter-process communication between child processes.
- Prevented zombie processes using `sigaction(SIGCHLD)` w/ non-blocking `waitpid()`, ensuring child process reaping.